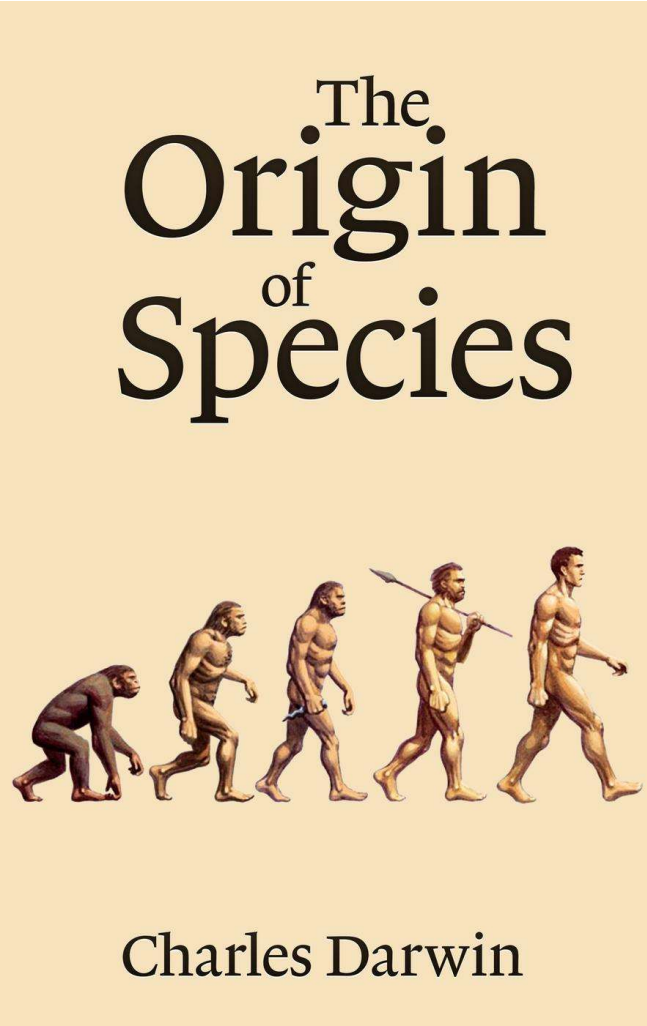
Speciation and Coevolution

The Origin of Species



Speciation - the origin of new <u>species</u> from pre-existing species.

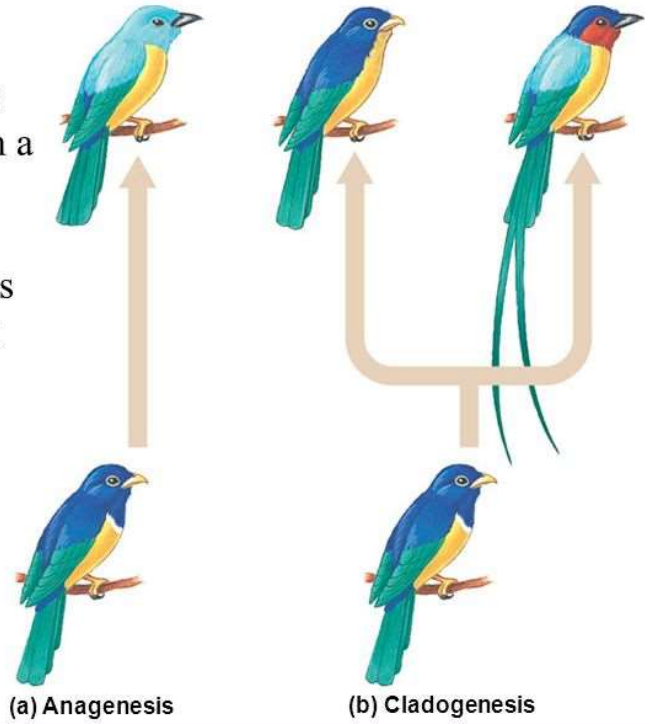
Certainly no clear line of demarcation has yet been drawn between species and sub-species – that is, the forms which…*come very near to, but do not quite arrive at, the rank of species*. …A well-marked variety may therefore be called an *incipient species*. …From these remarks it will be seen that I look at the term species as one arbitrarily given.

Darwin, *The Origin of Species*

Two patterns of evolutionary change

Anagenesis is evolution within a lineage.

Cladogenesis is evolution that results in the splitting of a lineage.



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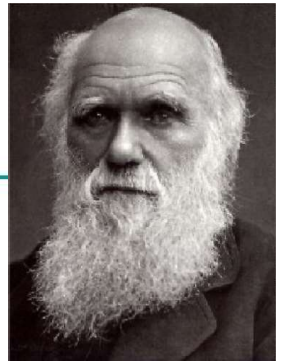

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"I was much struck how entirely vague and arbitrary is the distinction between species and varieties"
—*Charles Darwin, On the Origin of Species*

MAJOR ISSUES IN SPECIATION THEORY

- What is a species?
- How does speciation occur?
  - The role of geographic isolation in divergence.
  - Do population bottlenecks facilitate changes in genetic architecture?
  - The role of natural selection in speciation.
  - Prezygotic vs. postzygotic reproductive isolation.
  - The genetic mechanisms responsible for reproductive isolation.

The mind of the species problem

Jody Hey

Box 1. Species concepts[a]

- Agamospecies Concept
- Biological Species Concept*
- Cladistic Species Concept
- Cohesion Species Concept*
- Composite Species Concept
- Ecological Species Concept*
- Evolutionary Significant Unit*
- Evolutionary Species Concept*
- Genealogical Concordance Concept
- Genetic Species Concept*
- Genotypic Cluster Concept
- Hennigian Species Concept*
- Internodal Species Concept
- Morphological Species Concept
- Non-dimensional Species Concept
- Phenetic Species Concept
- Phylogenetic Species Concept (Diagnosable Version)*
- Phylogenetic Species Concept (Monophyly Version)
- Phylogenetic Species Concept (Diagnosable and Monophyly Version)
- Polythetic Species Concept
- Recognition Species Concept*
- Reproductive Competition Concept*
- Successional Species Concept
- Taxonomic Species Concept

Reference

a Mayden, R.L. (1997) A hierarchy of species concepts: the denouement in the saga of the species problem. In *Species: the Units of Biodiversity* (Claridge, M.F. *et al.*, eds), pp. 381–424, Chapman & Hall

*Concepts that make reference to biological processes (e.g. reproduction and competition) that occur among organisms within species (and less so between species) and that contribute to a shared process of evolution within species.

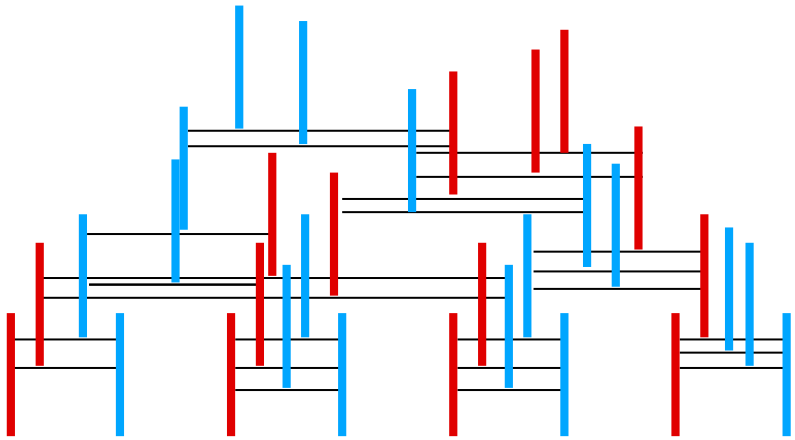
SPECIATION

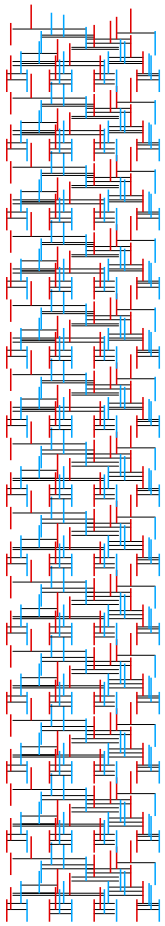
- The process by which one genetically-cohesive population splits into two or more reproductively-isolated populations.

- Requires the disruption of gene flow and the evolution of *reproductive isolating mechanisms* (RIMs).

Other species “concepts” exist

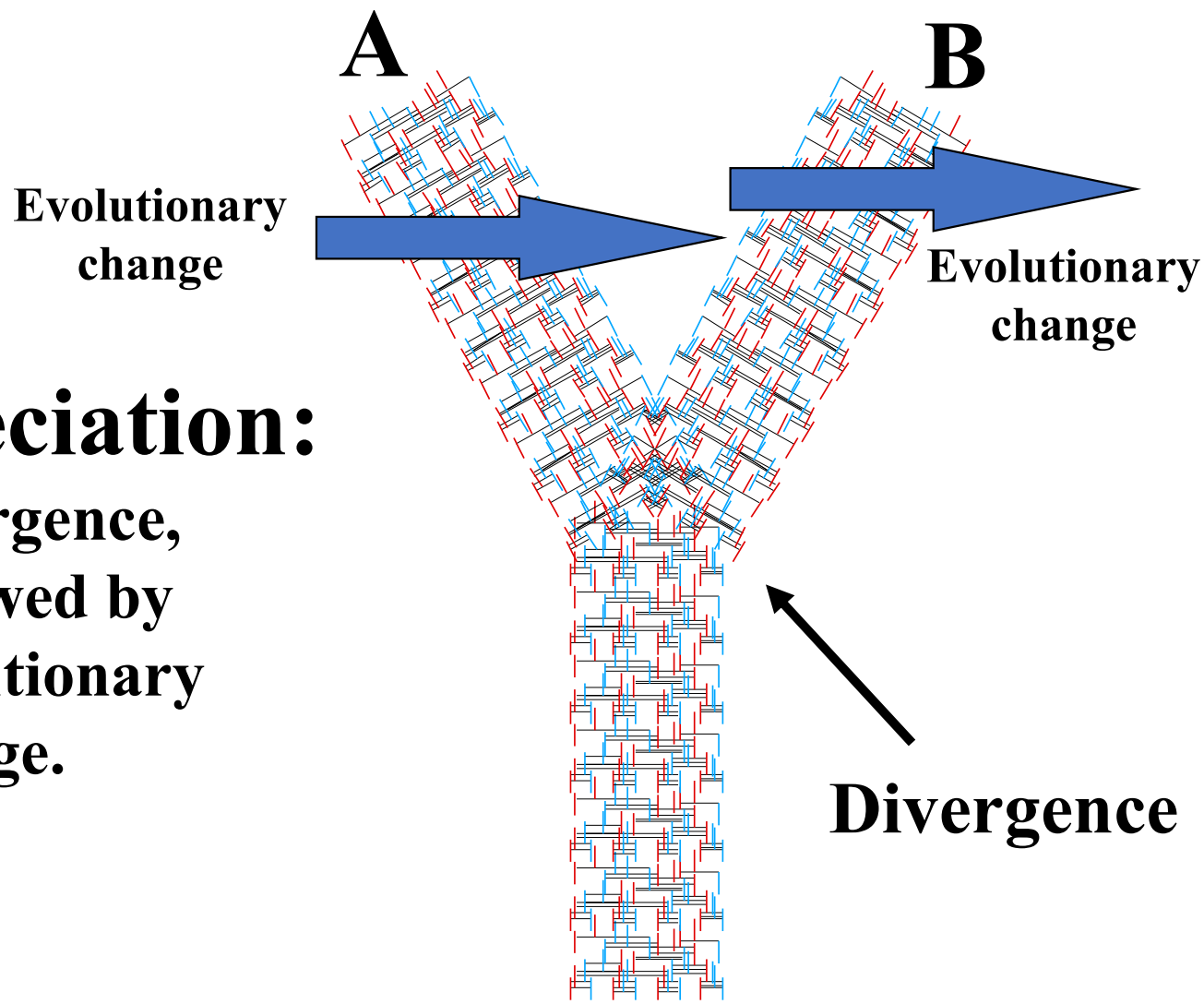
Interbreeding within species = lineage



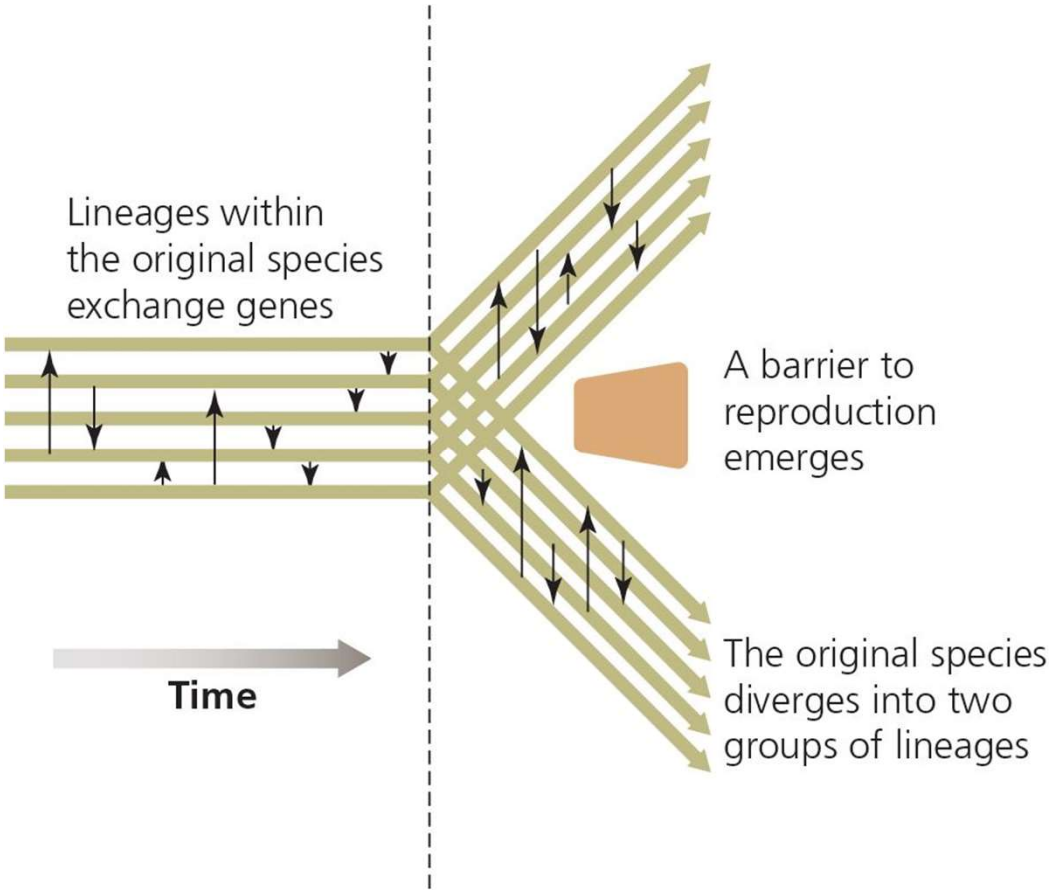


Speciation:

Divergence, followed by evolutionary change.



Speciation involves barriers to reproduction



What is a species?

(Latin for kind, type)

Biological Species:

= A set of naturally interbreeding populations that are genetically reproductively isolated from other sets of populations.

E. Mayr (1942)

PROBLEMS WITH THE BIOLOGICAL SPECIES CONCEPT

- Difficult to apply to fossils

- Asexual organisms don't fit the criteria

Types of Speciation

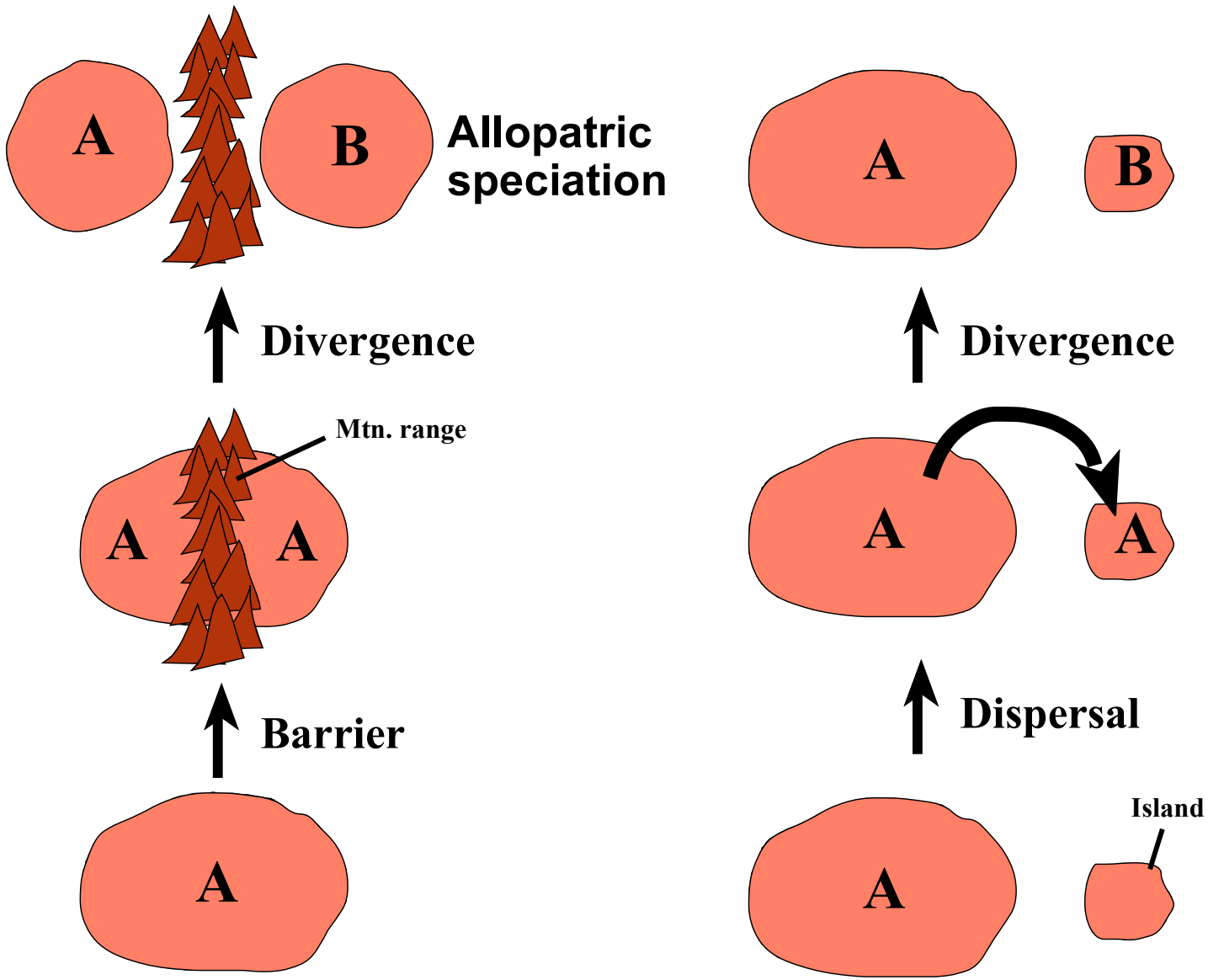
1) Allopatric

2) Sympatric

Allopatric speciation

= evolutionary change occurring in <u>different</u> geographic ranges.

Ancestral population divides;
each can undergo <u>independent</u> <u>evolutionary change</u>.


A
B
Allopatric speciation
Divergence
Mtn. range
A
A
Barrier
A
A
B
Divergence
A
A
Dispersal
Island
A

Sympatric speciation

= evolutionary divergence occurring in <u>same</u> (<u>overlapping</u>) geographic ranges.

Rare in nature,

but may occur by:

- **Initial disruptive selection (e.g., different food sources).**
- **Local ecological niche specialization (e.g., races/ecotypes)**

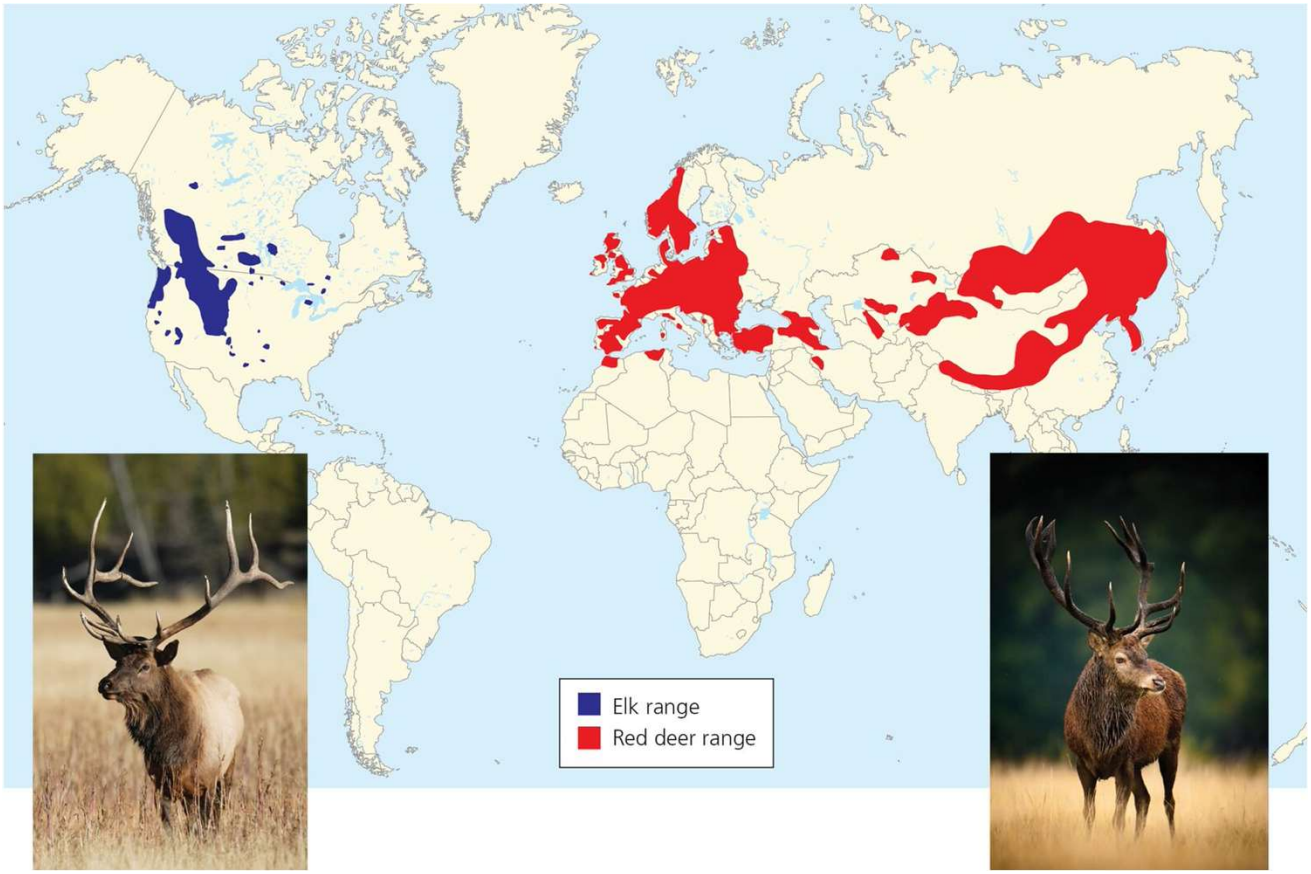
THE DIFFICULTY OF RECOGNIZING SPECIES

- For sympatric species, it usually is clear -- If they do not interbreed, then they are distinct species.
- For allopatric populations, it is less clear -- It may be difficult to assess whether they are "potentially interbreeding".
- It may also be difficult to decide what constitutes reproductive isolation.

  - If individuals from different populations produce sterile hybrids, then they are certainly reproductively isolated.

  - What if they produce partially sterile hybrids?

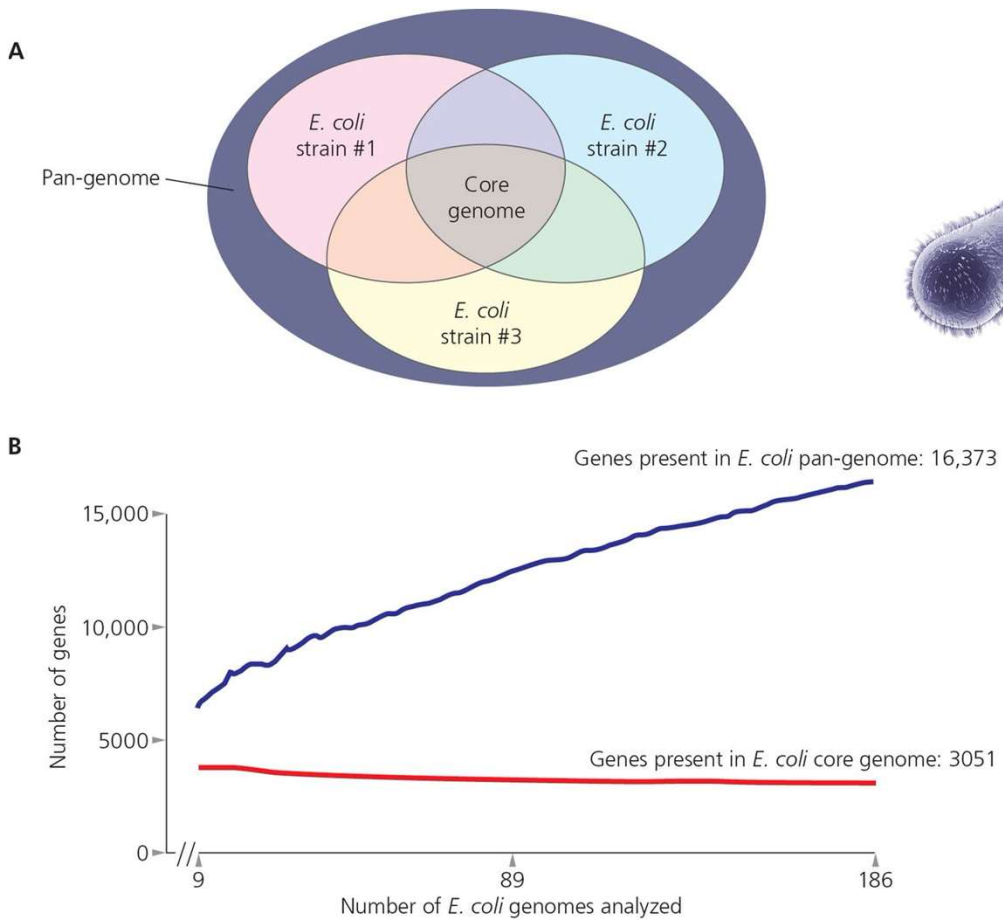
ELK AND RED DEER: TWO SPECIES?

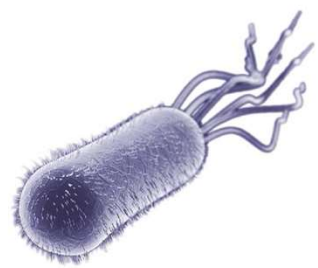


OTHER SPECIES CONCEPTS

- **PHYLOGENETIC SPECIES CONCEPT:** smallest possible group descending from a common ancestor and diagnosable by unique, derived traits
  - The PSC can be applied to virtually any taxonomic group including microbes and can be used on fossils.
  - However, at low levels of divergence (tips of the tree) it can be decoupled from reproductive isolation

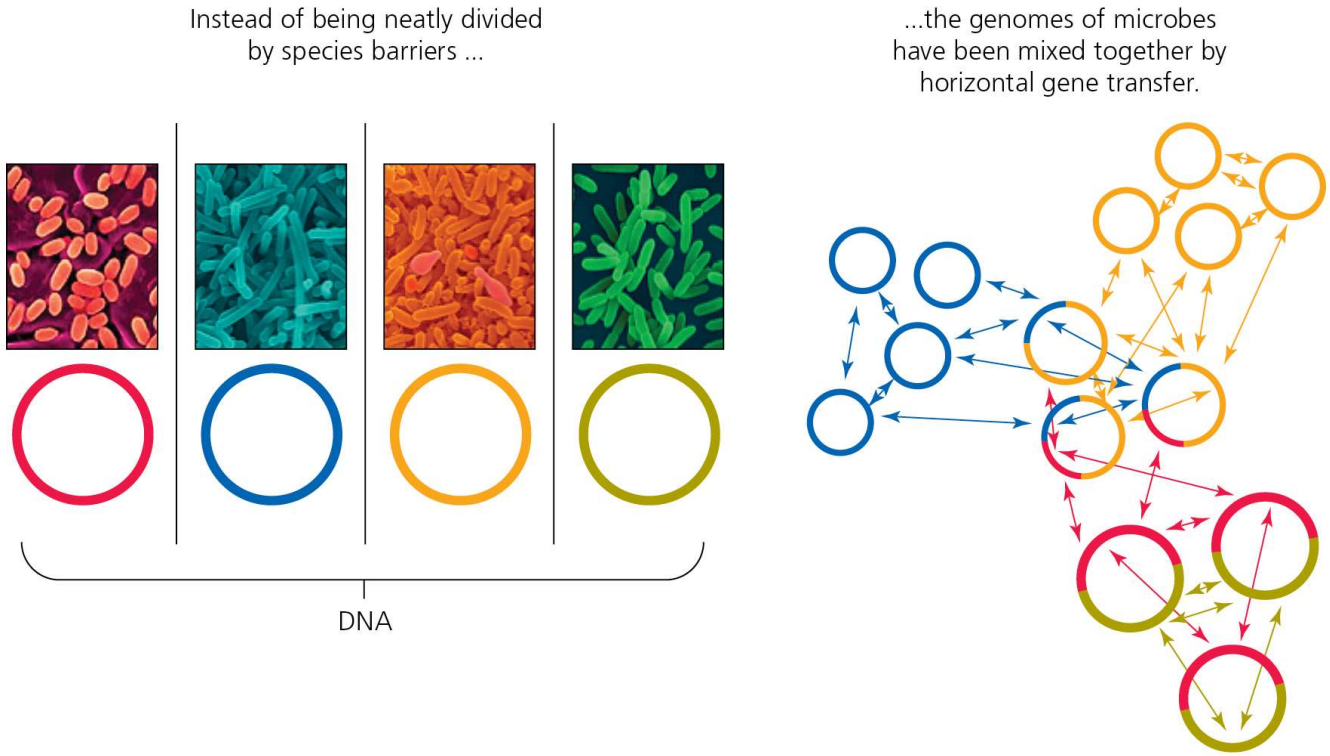
HORIZONTAL GENE TRANSFER IN *E. COLI*





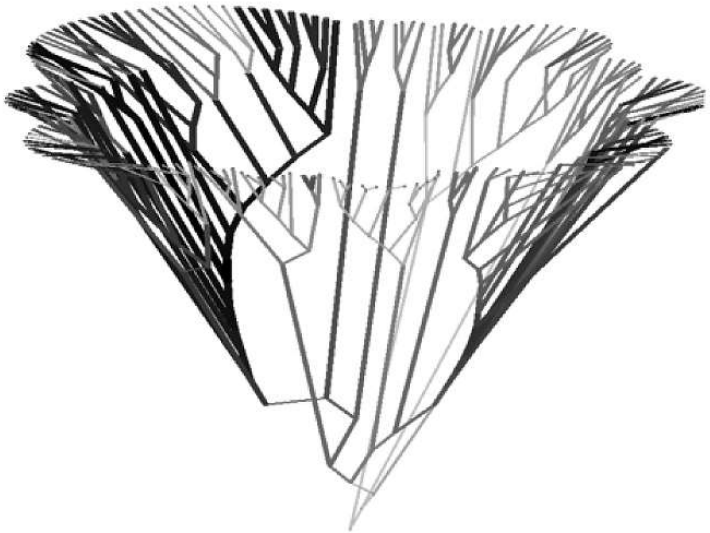
- **Only ~6% of the genes are shared by all strains (Kaas *et al.* 2012)**

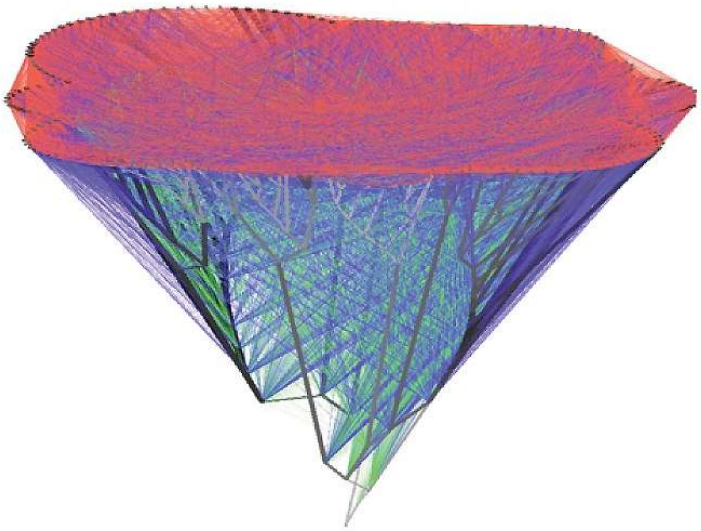
HORIZONTAL GENE TRANSFER MAKES CLASSIFICATION DIFFICULT



- **From an operational standpoint microbial species are often determined by a 97% similarity cut-off. (Yikes!)**

HORIZONTAL GENE TRANSFER IS COMMON IN MICROBIAL SYSTEMS





- How does this transfer of genetic material impact our views on common ancestry and phylogenetic relationships?

KEY FEATURES OF OPERATIONAL SPECIES CONCEPTS

- **Reproductive cohesion within species.**
- **Reproductive isolation from other such groups.**
- **Recognition that species are dynamic evolutionary lineages, not static “types”.**

OPERATIONAL SPECIES CONCEPTS PROVIDE A MEANS TO PROTECT:

- Individual populations
- Threatened portions of species ranges
- Ecologically distinct populations

Example: Separate runs of salmon

However, there are still problems…

What do we do with ***Hybrid Taxa***?

Red Wolf

ENDANGERED SPECIES

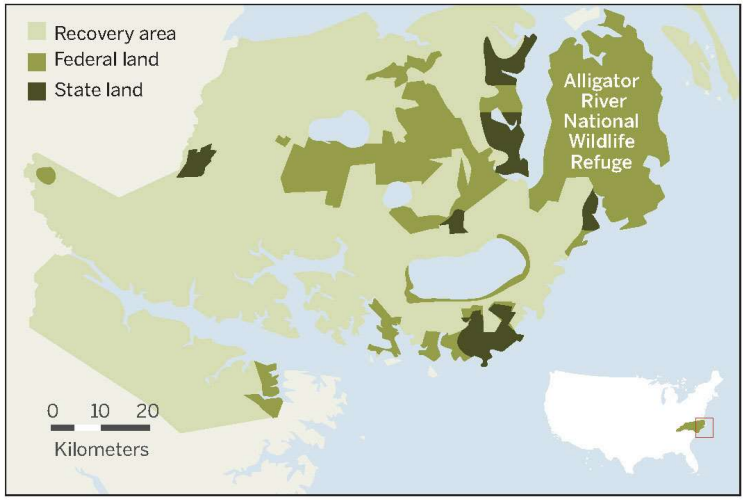
Red wolves in the crosshairs

U.S. agency ponders future of innovative reintroduction as animal deaths and controversy mount

Red wolves reproduce in the wild only in North Carolina. The population is increased with pups born in captivity. A major threat to the population is hybridization with coyotes.

Wild predators and private property

Red wolf recovery area includes land held by owners opposed to reintroduction



HYBRID ZONES

- *Hybridization* refers to crosses between genetically differentiated forms.
- *Introgression* refers to the movement of genes between species (or between well-differentiated populations) mediated by backcrossing.
- Hybridization is common. Thousands of examples have been documented in animals, tens of thousands in plants.

CLOSELY RELATED SPECIES OFTEN EXHIBIT HYBRID ZONES WHEN THEY EXPERIENCE SECONDARY CONTACT

Corvus cornix

Corvus corone

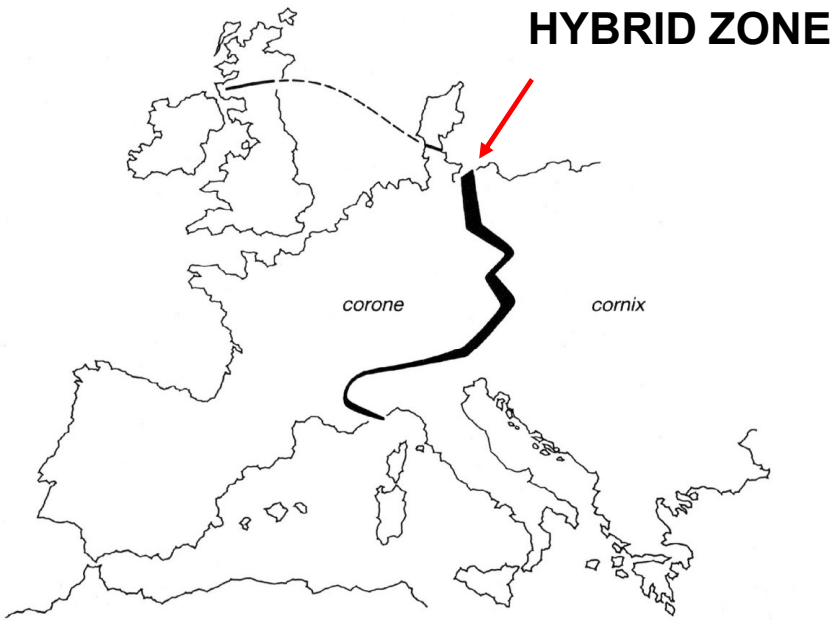


Figure 16.8 Hybrid zone between the hooded crow (*Corvus corone*) and carrion crow (*C. cornix*) in Europe. From Mayr (1963).